

University of Khartoum
Faculty of Engineering
DBS Exam Answers

Question ①:

NO	Answer
1	(C) DCL
2	(D) None of the above
3	(C) Doubly outlined rectangle
4	(B) Constraints
5	(B) Foreign Key
6	(D) SELECT
7	(A) To retrieve unique values from a column
8	(A) Having
9	(b) Group By clause
10	(A) INNER JOIN
11	(C) Where
12	(B) Cartesian product
13	(C) commit
14	(C) In-Memory Database Management System (IMDBMS)
15	(A) RAID 1+0
16	(B) Buffer manager
17	(B) Least recently used (LRU)
18	(C) Primary Key
19	(c) Right
20	(C) B+ tree allows rapid random access as well as rapid sequential access

Question(2):

1. Four components of a DBMS:

- A _ Query processor / query compiler
- B_ Transaction manager
- C_ Storage manager
- D_ Logging and recovery manager

2. Describe the relational model:

The relational model represents data as two-dimensional tables called relations. Columns are called attributes, and rows are called tuples. The relation name with its attributes forms the schema. Each attribute has a domain, and every tuple component must be atomic. Operations on relations are defined by relational algebra.

3. Referential integrity constraint A constraint ensuring that a foreign key value in one relation must match an existing primary key value in a related relation, so references between tables remain valid.

4. ACID properties

- Atomicity — all-or-nothing execution.
- Consistency — moves the database between valid states.
- Isolation — concurrent transactions do not interfere with each other.
- Durability — committed changes persist even after a failure.

5. Two data pipeline patterns

- ETL (Extract, Transform, Load)
- ELT (Extract, Load, Transform)

6. Storage hierarchy Cache → Main memory → Flash storage → Magnetic disk → Optical disk/Tape. Moving down the hierarchy, access time increases, cost per unit decreases, and capacity increases. Frequently used data is kept higher; large infrequently used data is kept lower.

7. RAID level with least rebuild interference RAID 1 (mirroring). Rebuilding only copies data from the surviving mirror disk, unlike parity-based RAID levels (5, 6) which must read all disks and recompute data via XOR.

8. • Ordered indices — store search keys in sorted order; support point and range queries. • Hash indices — map keys to buckets via a hash function; fast for point queries only, no range query support.

Basis for choosing an ordered technique: access type, access time, insertion time, deletion time, space overhead.

9. Multi-level indexing An index built over another index (outer index over inner index), used when a single-level index becomes too large to search efficiently, reducing the number of block accesses.
10. Stored procedures A stored procedure is a named, precompiled block of SQL statements stored and executed on the database server. Advantages: reduces network traffic, promotes code reuse, improves performance, enhances security.
Creation:
`CREATE PROCEDURE procedure_name (parameters) BEGIN – SQL statements
END;`
11. Why concurrency control is needed To prevent problems from uncontrolled interleaved execution: lost update, temporary update (dirty read), incorrect summary, and unrepeatable read. It ensures isolation and consistency, equivalent to a serial execution.

Question ③:

1. 7 tuples
2. The number of authors who have never been an editor of any book
- 3.

$$\bullet \pi_{first_name, last_name} (author \bowtie_{author_id=editor} book)$$

- 4.

$$\sigma_{first_name \geq 'A' \wedge first_name < 'B'} (author)$$

Question ④:

--1. Create tables

```
CREATE TABLE Department ( DepID INT PRIMARY KEY,  
    Dname VARCHAR(50) NOT NULL,  
    MgrID INT );  
CREATE TABLE Employee (  
    EmpID INT PRIMARY KEY,  
    Fname VARCHAR(50) NOT NULL,  
    Lname VARCHAR(50) NOT NULL,  
    Bdate DATE, Gender CHAR(1),  
    Salary DECIMAL(10,2),  
    SupervisorID INT,  
    Dno INT,  
    FOREIGN KEY (SupervisorID) REFERENCES Employee(EmpID),  
    FOREIGN KEY (Dno) REFERENCES Department(DepID) );  
ALTER TABLE Department  
ADD FOREIGN KEY (MgrID) REFERENCES Employee(EmpID);
```

--2. Insert EmpID = 1

```
INSERT INTO Employee (EmpID, Fname, Lname, Bdate, Gender, Salary,  
    SupervisorID, Dno)  
VALUES (1, 'John', 'Smith', '1965-01-09', 'M', 30000.00, 2, 5);
```

--3. Total number of employees

```
SELECT COUNT(*) AS Total_Employees FROM Employee;
```

--4. Employees in the Research department

```
SELECT E.*  
FROM Employee E JOIN Department D ON E.Dno = D.DepID WHERE  
D.Dname = 'Research';
```

--5. Average salary per department

```
SELECT Dno, AVG(Salary) AS Average_Salary FROM Employee GROUP BY  
Dno;
```

--6. Employees with department name

```
SELECT E.Fname, E.Lname, D.Dname FROM Employee E JOIN Department  
D ON E.Dno = D.DepID;
```